AI ASSISTANT CODING ASSIGNMENT-14.2

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Batch: 33

Semester: VI

Branch: CSE(AIML)

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation
Lab Objectives

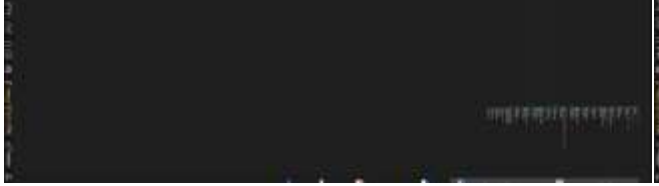
- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.
- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

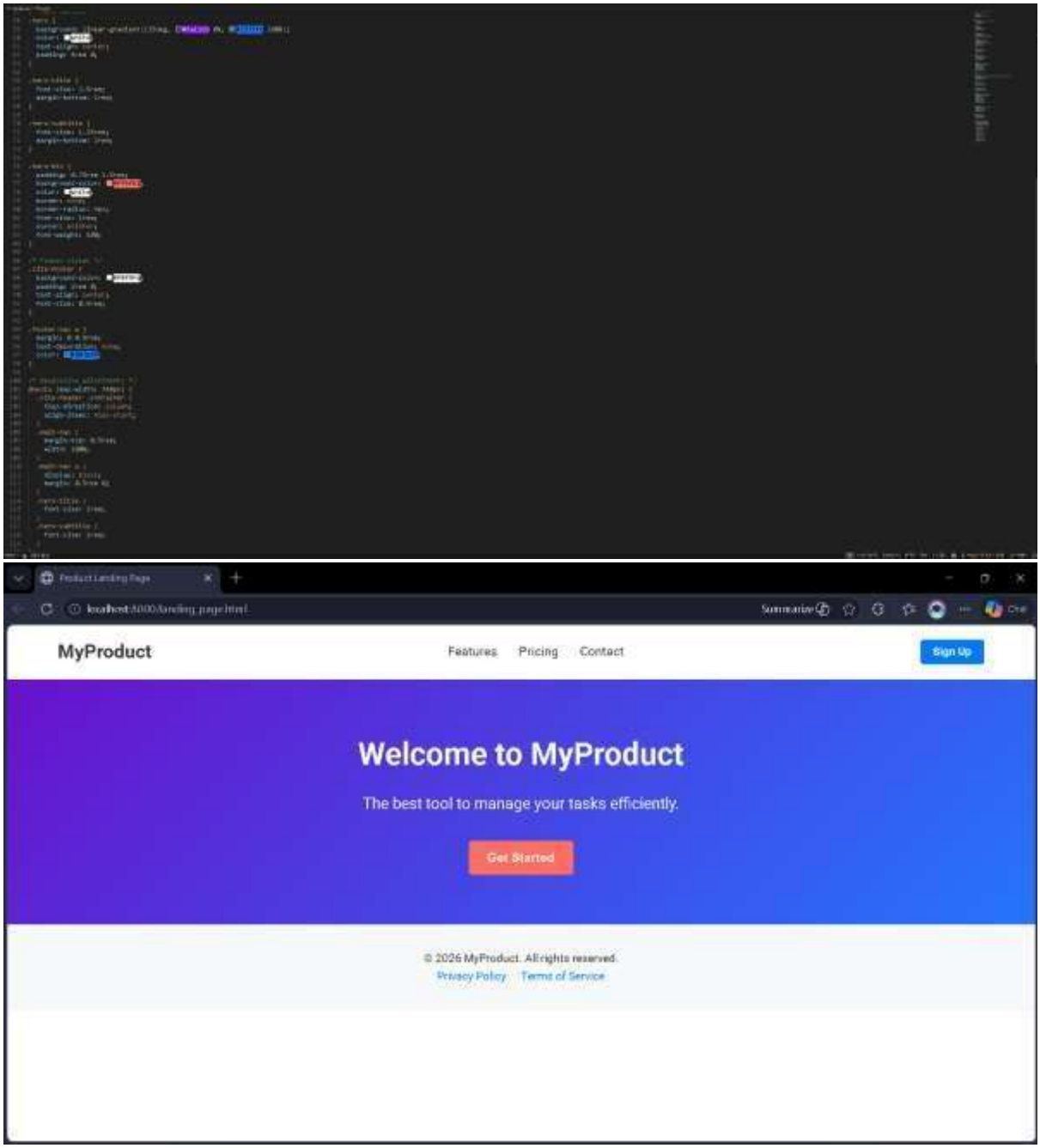
Task Description 1 – (Landing Page Design)

- Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.





Task Description 2 – (Theme Toggle Feature)

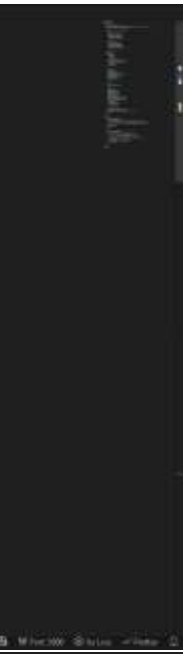
• Provide a button to switch between light and dark themes..

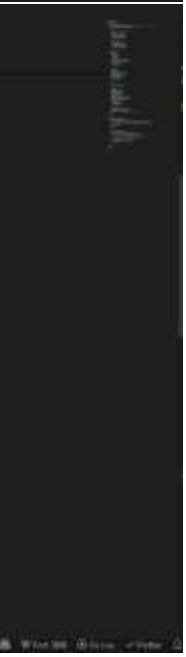
Requirements:

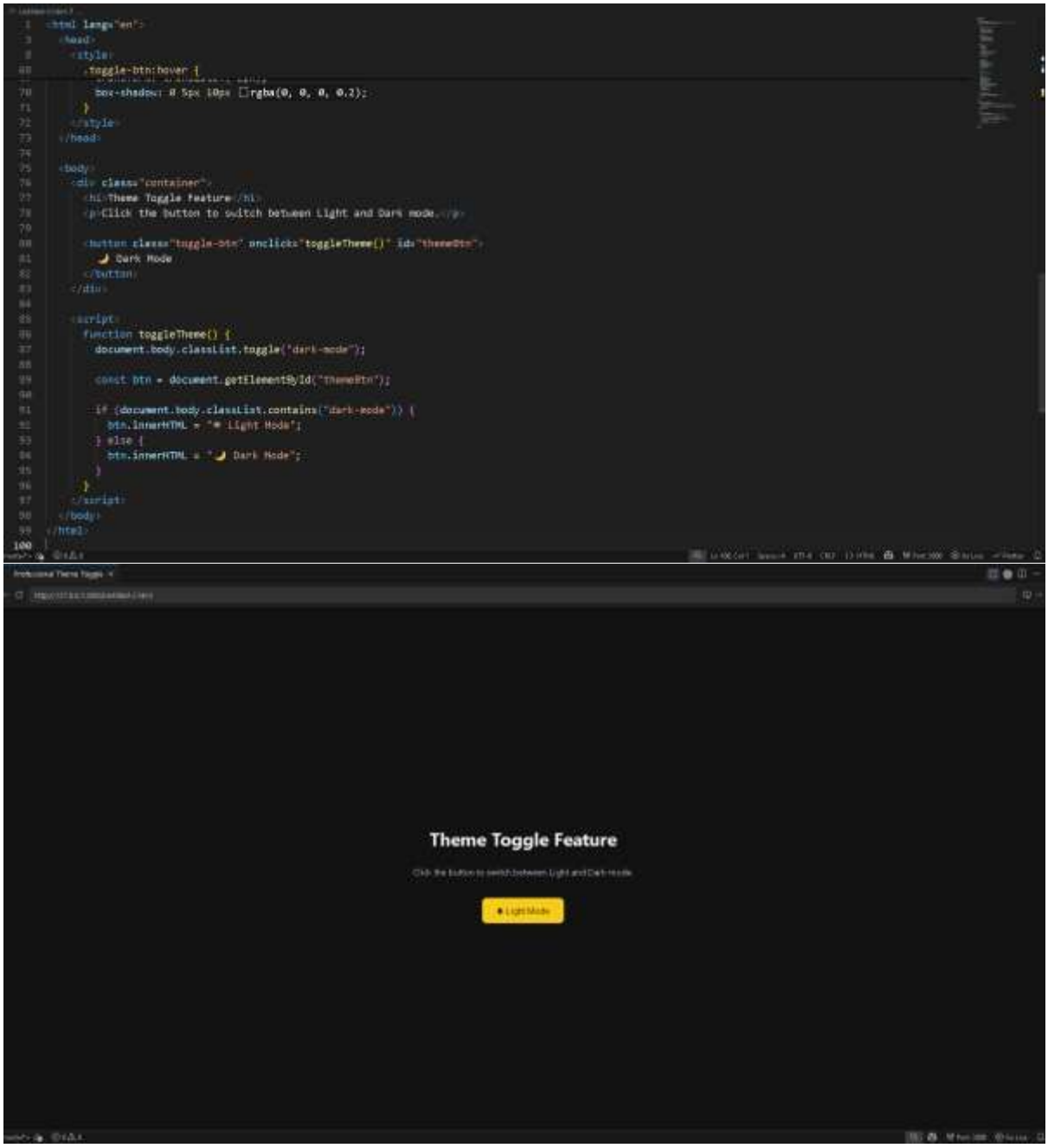
• Use JavaScript to toggle CSS classes.

• Use AI assistance to manage color contrast and accessibility.

• Ensure theme preference applies instantly on click.







Task Description 3 – (User Feedback Form)

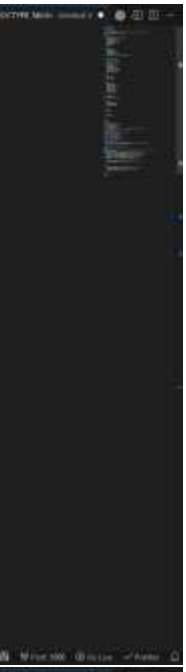
• Create a feedback form with fields: Username, Rating, Comments.

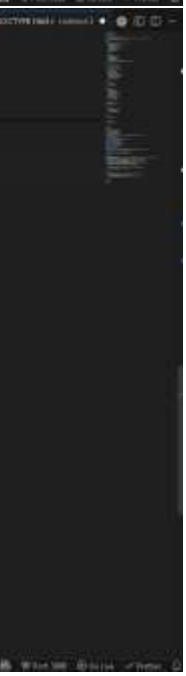
Requirements:

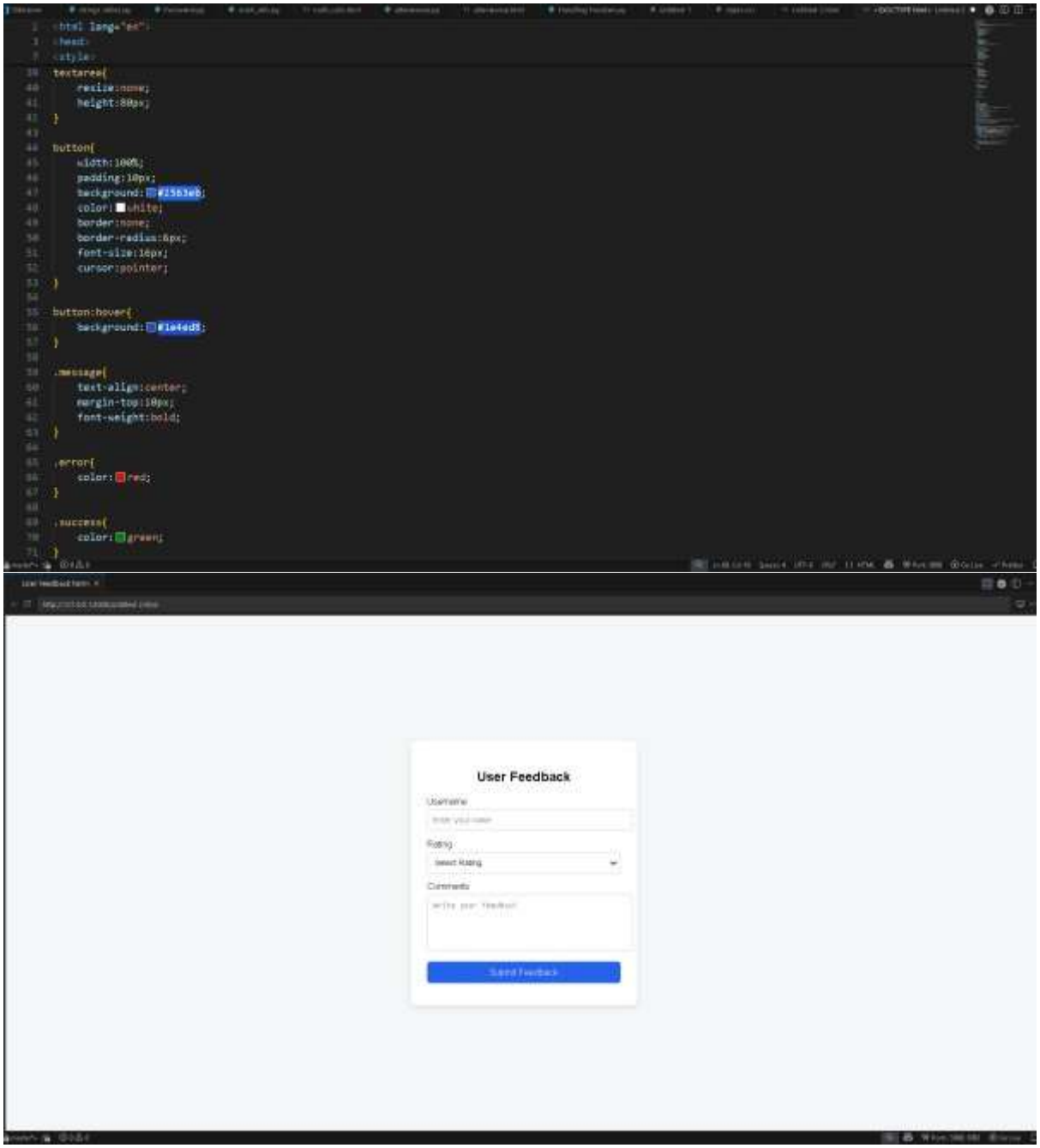
• Use JavaScript to validate required fields.

• Display success or error messages dynamically.

• Use AI to improve form usability and layout.







Task Description - 4 (Image Gallery with JavaScript)

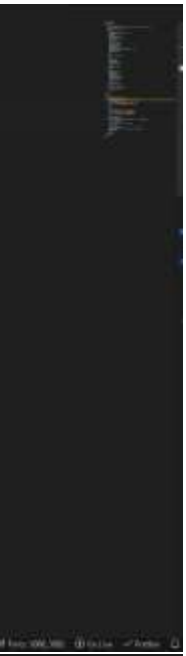
• Task: Display a small image gallery on a web page.

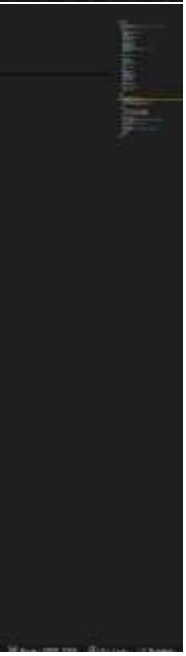
Requirements:

• Use JavaScript to dynamically load images.

• Add next/previous navigation buttons.

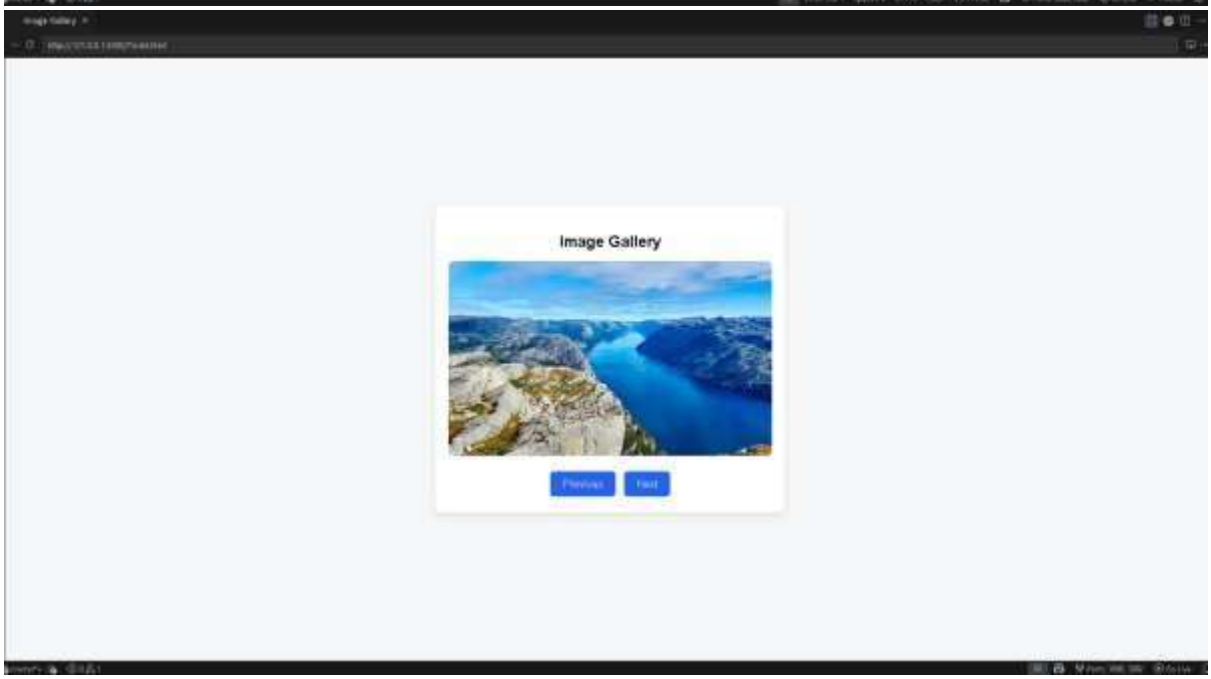
• Use AI to optimize layout responsiveness.





```html
<html lang="en">
  <head>
  <body>
    <div class="gallery-container">
      <h2>Image Gallery</h2>
      <img id="galleryImage" src="" alt="Gallery Image" />
      <div class="buttons">
        <button onclick="prevImage()">Previous</button>
        <button onclick="nextImage()">Next</button>
      </div>
    </div>
    <script>
      const images = [
        "https://picsum.photos/id/1015/600/400",
        "https://picsum.photos/id/1025/600/400",
        "https://picsum.photos/id/1035/600/400",
        "https://picsum.photos/id/1045/600/400",
      ];
      let index = 0;
      function showImage() {
        document.getElementById("galleryImage").src = images[index];
      }
      function nextImage() {
        index = (index + 1) % images.length;
        showImage();
      }
      function prevImage() {
        index = (index - 1 + images.length) % images.length;
        showImage();
      }

      showImage();
    </script>
  </body>
</html>
```

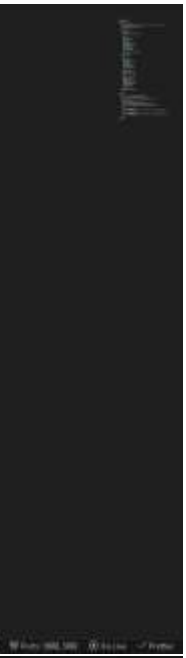


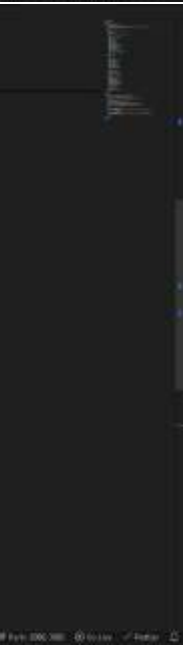
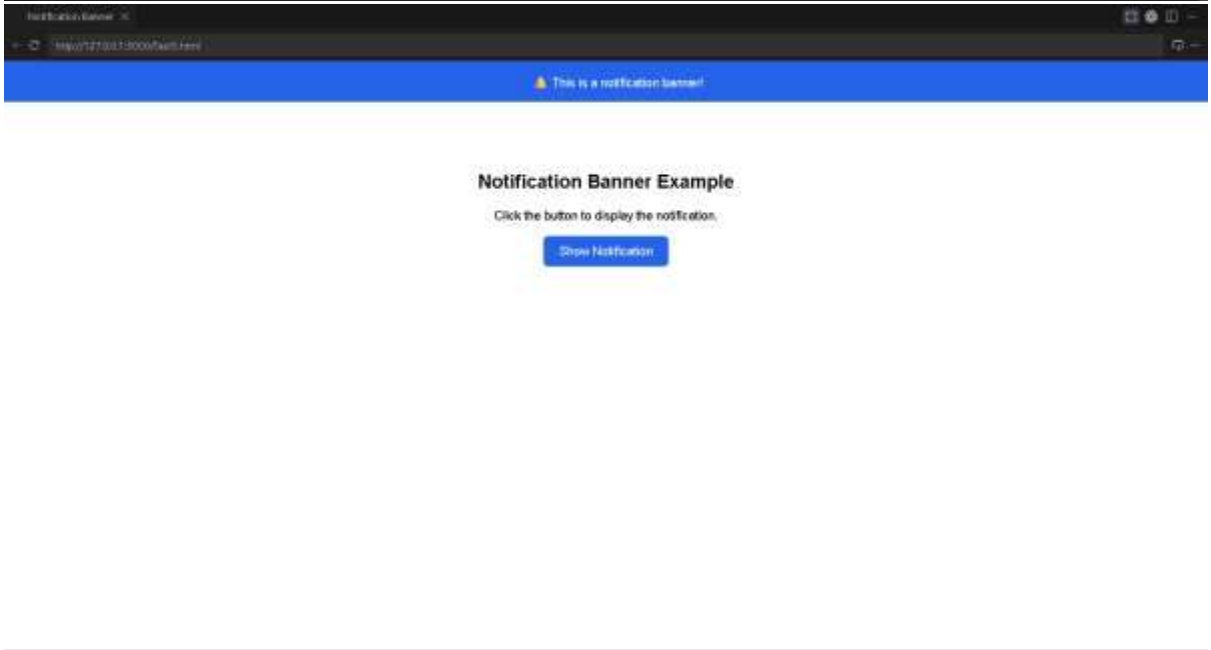
Task Description - 5: (Notification Banner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.





```html
    <script>
      function showBanner() {
        document.getElementById("notificationBanner").classList.add("show");
      }
      function hideBanner() {
        document.getElementById("notificationBanner").classList.remove("show");
      }
    </script>
  </body>
</html>
```